MapCSS

CSS-like map stylesheets

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Mapnik is the great renderer but it stylesheets scares me:

- XML
- 416 Kb
- 9536 Lines of code
- Mixed declarative selectors and SQL

```
<Layer name="tracks-tunnels" status="on" srs="&osm2pgsql_projection;">
   <StyleName>tracks-tunnels</StyleName>
   <Datasource>
      <Parameter name="table">
(select way,tracktype from &prefix;_line where highway='track' and tunnel in ('yes','true','1')) as tracks
      </Parameter>
&datasource-settings;
   </Datasource>
</Layer>
```

Why MapCSS?

- Simple (same terms as in OSM)
- Lightweight (small files)
- Declarative (*what* instead of *how*)
- Map Styling language



How it's looks like?

```
line[highway=primary],
line[highway=secondary] {
    width: 5;
    color: #E77817;
    casing-width: 5;
    casing-color: #E77817;
    text: name;
    font-size: 12;
    text-position: line;
}
line[waterway] {
    width: 5;
    color: blue;
}
area[building] {
    fill-color: grey;
    text: addr:housenumber;
}

@import url("park.mapcss")
```



Style anatomy

```
line[highway=primary]  /* select non-closed lines */
{
    width: 3;
    color: gray;
    /* Render as  3 pixels gray line */
}

area[building]   /* select closed ways with and building tag */
{
    fill-color: #500; /* Polygon fill color */
}

node|z11-14[place][population>5000],
area|z11-14[place][population>1000]
{
    text: name;
}
```

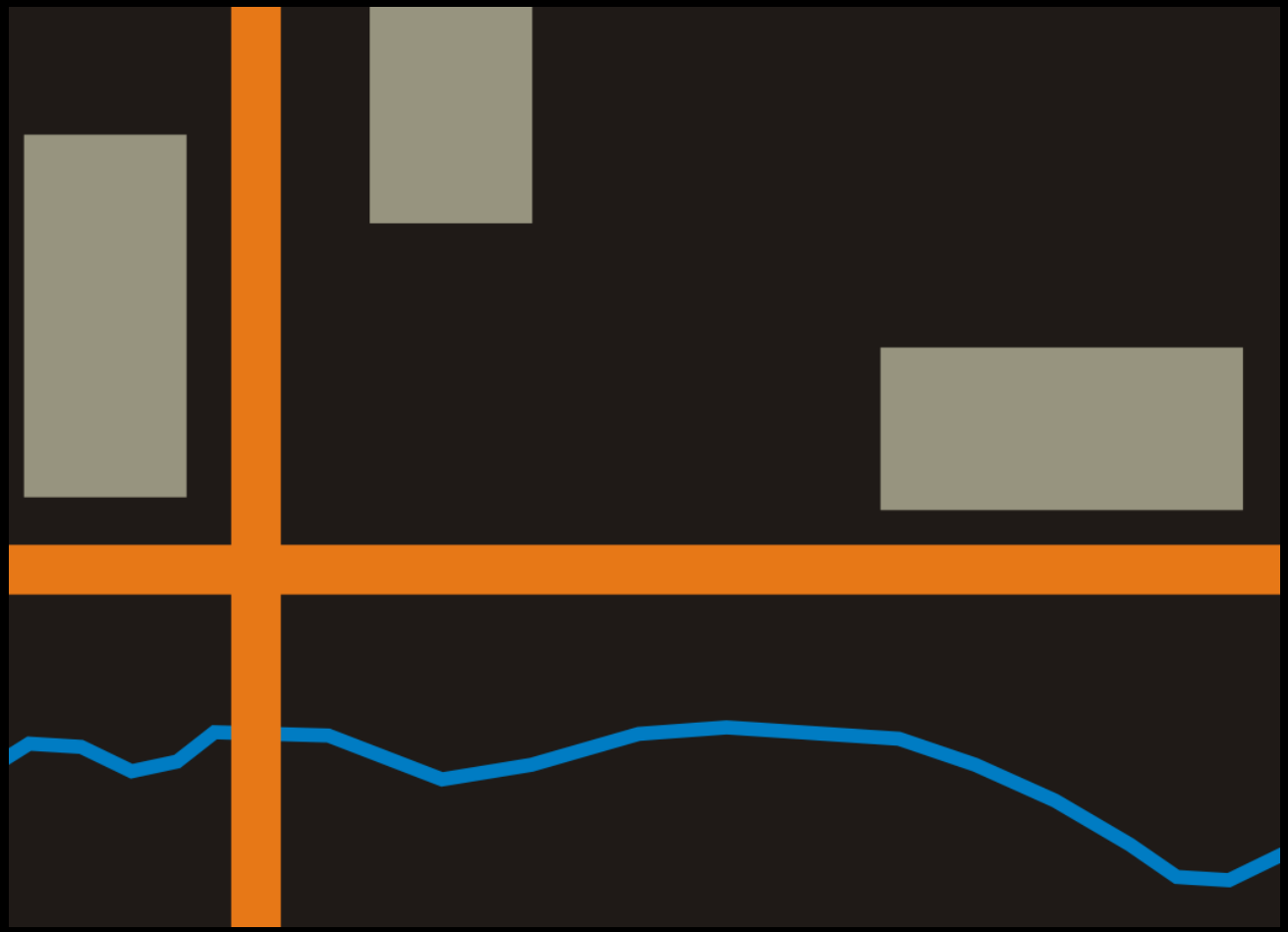
Selectors

node|z11-14[place][population>5000],
area|z11-14[place][population>1000]

- Elements (node, way, relation, line, area)
- Zoom node|z14-
- Tag value [highway=primary]
- Tag presence [bridge] (any value of bridge tag)
- Range [population>50000]
- Regular expression [highway=~/^primary/]
- AND/OR chaining

Simple style

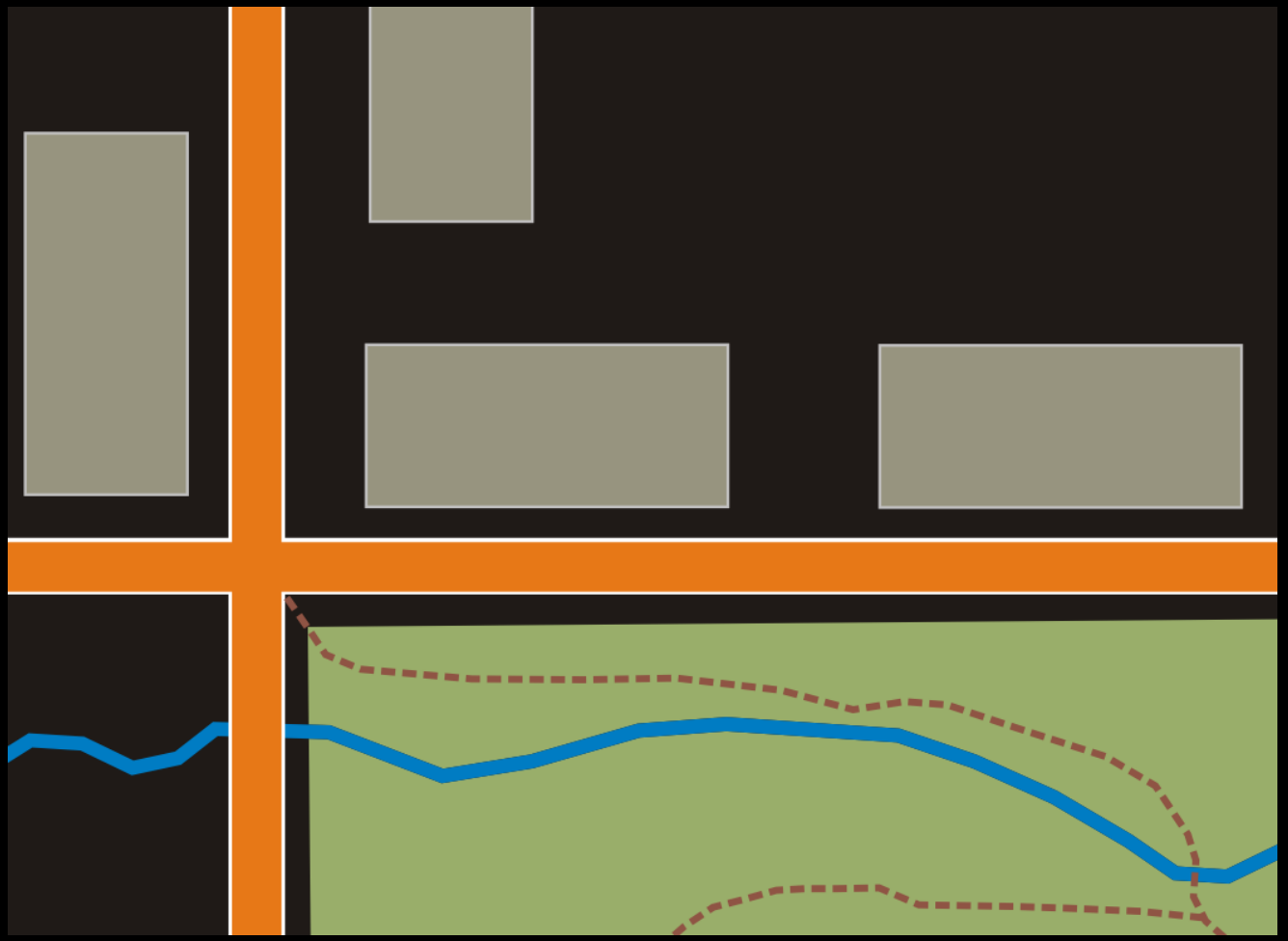
```
line[highway] {
  color: orange;
  width: 11;
}
area[building] {
  fill-color: gray;
}
line[waterway] {
  color: blue;
  width: 3;
}
```



Make map looks better

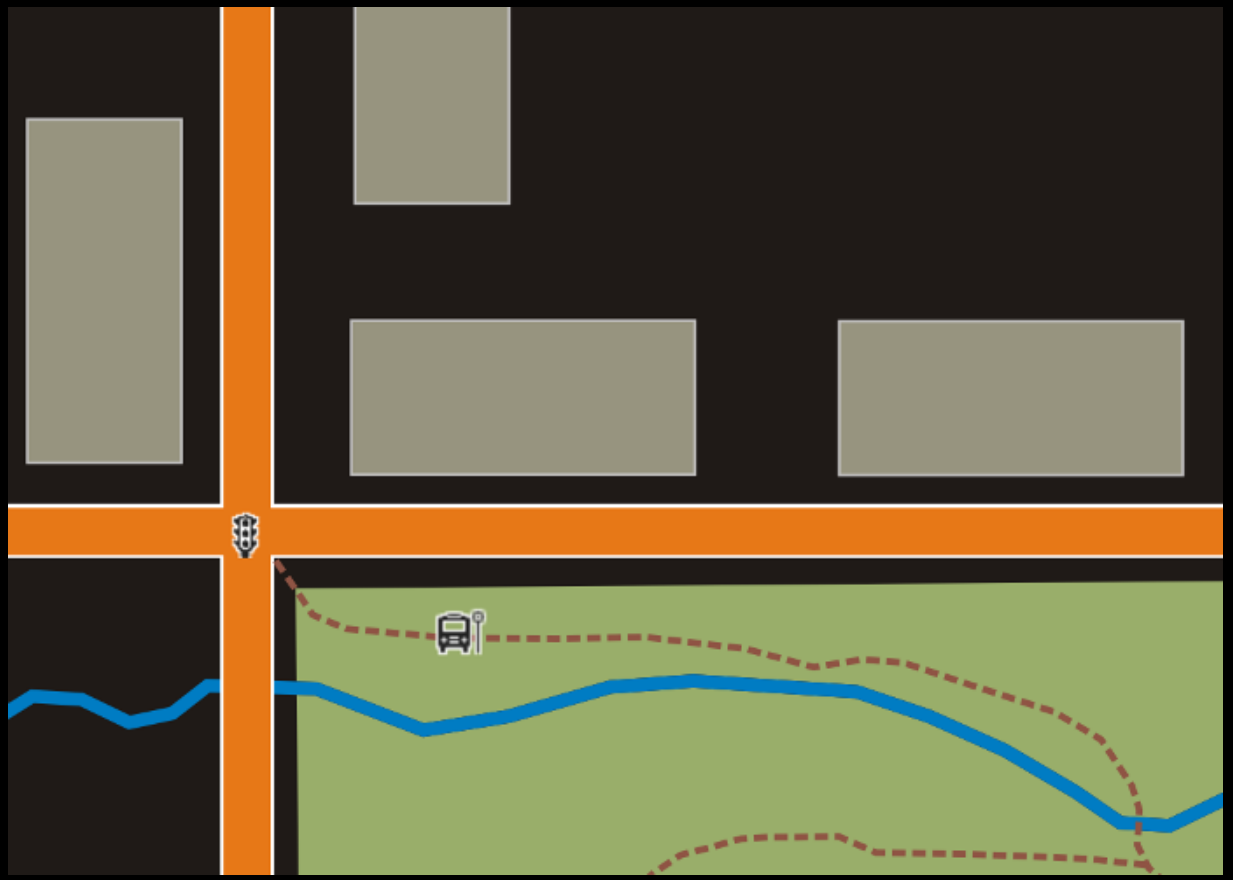
```
line[highway=primary] {
  casing-color: white;
  casing-width: 12;
  casing-opacity: 0.5;
}

line[highway=track] {
  color: brown;
  width: 1;
  dashes: 3,3;
}
```



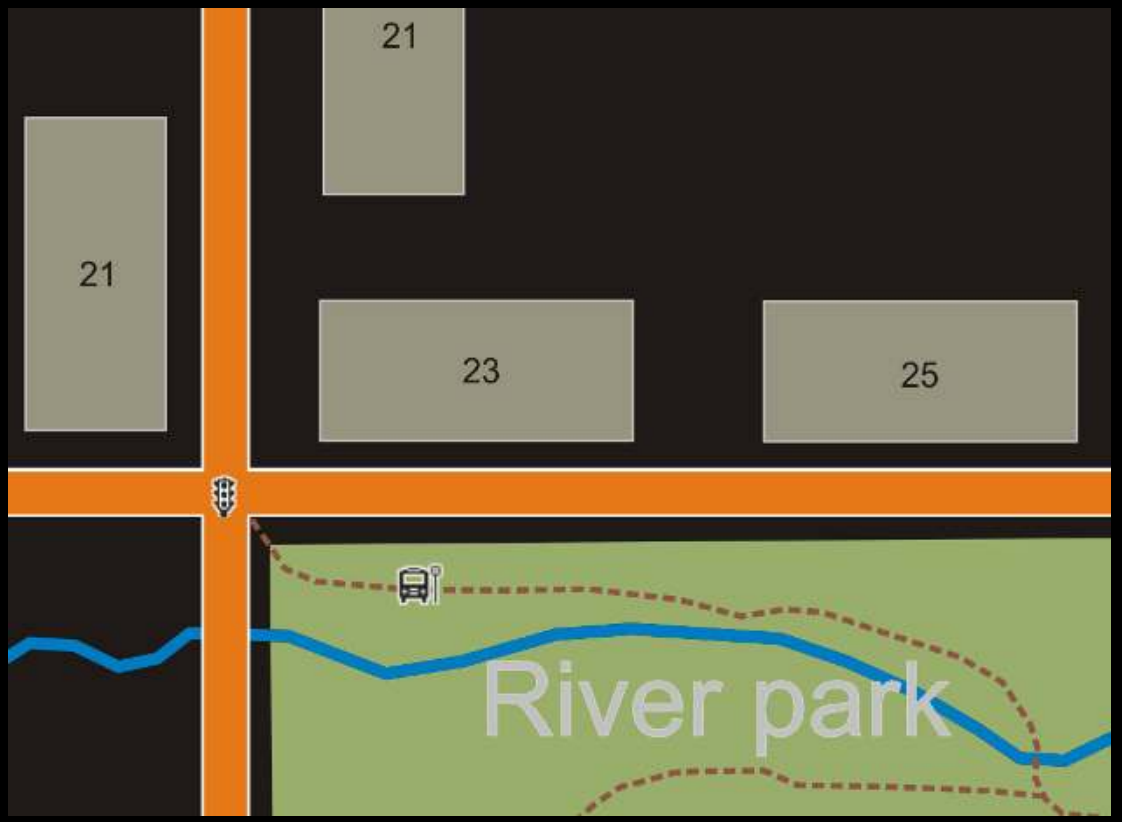
Icons

```
node[highway=bus_stop] {
  icon-image: bus_stop.png;
}
node[highway=traffic_lights] {
  icon-image: url("http://example.com/traffic_lights.png");
}
```



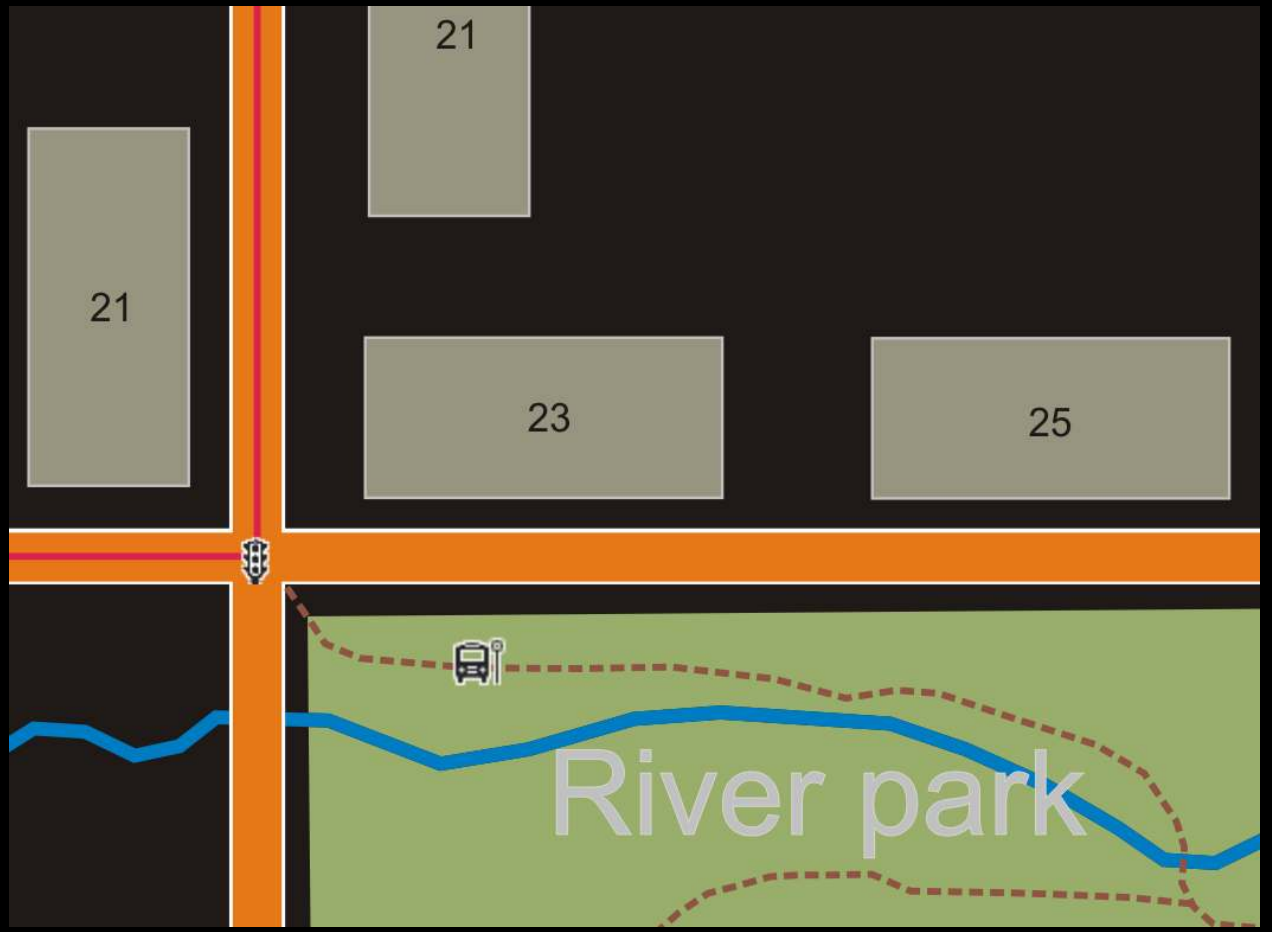
Labels

```
node[name] {
  text: name;
  font-size: 20;
  text-color: grey;
  text-halo-color: lightgrey;
  text-halo-radius: 1px;
}
area[addr:housenumber] {
  text: addr:housenumber;
  font-size: 11;
  text-color: black;
  text-position: center;
}
```



Subparts

```
line[railway=tram][highway]::tram {
  width: 1;
  color: red;
  z-index: 10;
}
```



Eval is not evil

```
line[highway=service] {
  width: eval(min(10, any(metric(tag("width")), tag("lanes") * 5)));
}
```

- Built-in Javascript-like DSL
- Simple arithmetic operators
- Few useful functions (min, max, any, cond)
- Special metric() magic converts tag value to pixels

A silver bullet?

- No complex vector symbols
- Hard to maintain multi-scale styles
- Cannot change symbol orientation
- No tools support

How to use Map CSS

- Renderers: Halcyon, Ceyx, Kothic, Kothic-JS
- Editors: JOSM, Potlatch 2
- Komap (Mapnik style preprocessor)
- MapCSS parser

What's next?

- Current version is 0.2
- Wiki page
- Mailing list at mapcss@openstreetmap.org
- #osm IRC channel at OFTC

MapCSS

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Q/A

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